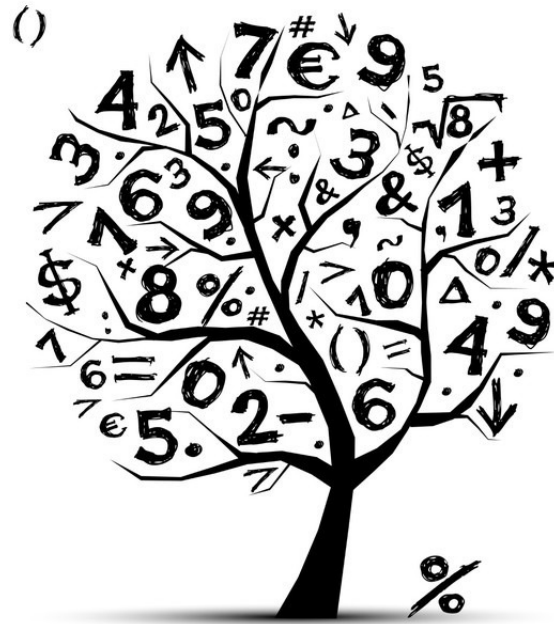


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Sums



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1 Definition

Definition 1: sum

Given n values $a_j \in \mathbb{R}$ with $j \in \{1, 2, \dots, n\}$, the **sum**

$$a_1 + a_2 + \dots + a_n$$

can be indicated in compact form with the sum symbol:

$$\sum_{j=1}^n a_j$$

which reads: “sum for j from 1 to n of a_j ”. The symbol j is called **sum index**.

- The sum symbol is therefore very useful when the terms a_j are explicitly defined as a function of the sum index j .

Example 1: sums

$$\sum_{j=1}^{10} \frac{1}{j} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{j=3}^n j^2 = 3^2 + 4^2 + 5^2 + \dots + n^2$$

- The sum index is a **dummy index**. This means that if j is replaced with i , k or any other index in all its occurrences the value of the sum does not change.

Example 2: dummy index

We have:

$$\sum_{j=1}^n j^2 = \sum_{i=1}^n i^2 = \sum_{k=1}^n k^2$$

as the two symbols indicate the sum of the squares of the first n natural numbers (without zero). On the contrary, we have:

$$\sum_{j=1}^n j^2 \neq \sum_{j=1}^m j^2$$

as the two symbols indicate the sum, respectively, of the first n and the first m squares of the natural numbers (without zero). Clearly, if $n \neq m$, the result is different.

2 Properties

Observation 1: sum with constant term

For every natural number $n \geq 1$ and real number c , we have:

$$\sum_{j=1}^n c = n c \quad (1)$$

Proof. We have:

$$\underbrace{c + c + \dots + c}_{= \sum_{j=1}^n c, \text{ } c \text{ summed up } n \text{ times}} = n c$$

□

Observation 2: product by a constant

For every sum $\sum_{j=1}^n a_j$ and real number c , we have:

$$\sum_{j=1}^n (c a_j) = c \sum_{j=1}^n a_j \quad (2)$$

Proof. From the distributive property, we have:

$$\underbrace{c a_1 + c a_2 + \dots + c a_n}_{= \sum_{j=1}^n (c a_j)} = \underbrace{c (a_1 + a_2 + \dots + a_n)}_{= c \sum_{j=1}^n a_j}$$

□

Observation 3: union of sums

For every pair of sums $\sum_{j=1}^n a_j$ and $\sum_{j=1}^n b_j$, we have:

$$\sum_{j=1}^n a_j + \sum_{j=1}^n b_j = \sum_{j=1}^n (a_j + b_j) \quad (3)$$

Proof. We have:

$$\underbrace{a_1 + a_2 + \dots + a_n + b_1 + b_2 + \dots + b_n}_{=\sum_{j=1}^n a_j + \sum_{j=1}^n b_j} = \underbrace{a_1 + b_1 + a_2 + b_2 + \dots + a_n + b_n}_{=\sum_{j=1}^n (a_j + b_j)}$$

□

Observation 4: decomposition

For every sum $\sum_{j=1}^{n+m} a_j$, we have:

$$\sum_{j=1}^{n+m} a_j = \sum_{j=1}^n a_j + \sum_{j=n+1}^{n+m} a_j \quad (4)$$

Observation 5: index translation

For every sum $\sum_{j=1}^n a_j$ and natural number $m \geq 1$, we have:

$$\sum_{j=1}^n a_j = \sum_{j=1+m}^{n+m} a_{j-m} = \sum_{j=1-m}^{n-m} a_{j+m} \quad (5)$$

Observation 6: index reflexion

For every sum $\sum_{j=1}^n a_j$, we have:

$$\sum_{j=1}^n a_j = \sum_{j=1}^n a_{n-j+1} = \sum_{j=0}^{n-1} a_{n-j} \quad (6)$$

Proof. The three properties are simply different notations and/or arrangements of the terms in the sums. □

3 Some important sums

Observation 7: sum of the first n positive natural numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=1}^n j = \frac{n^2 + n}{2} \quad (7)$$

Proof. We have:

$$\begin{aligned} \sum_{j=1}^n j &= \frac{1}{2} \left(\sum_{j=1}^n j + \sum_{j=1}^n j \right) \\ &= \frac{1}{2} \left(\sum_{j=1}^n j + \sum_{j=1}^n (n - j + 1) \right) \\ &= \frac{1}{2} \sum_{j=1}^n (j + n - j + 1) \\ &= \frac{1}{2} \sum_{j=1}^n (n + 1) = \frac{n(n+1)}{2} = \frac{n^2 + n}{2} \end{aligned}$$

□

Observation 8: sum of the first n positive even numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=1}^n 2j = n^2 + n \quad (8)$$

Proof. We have:

$$\sum_{j=1}^n 2j = 2 \sum_{j=1}^n j = 2 \left(\frac{n^2 + n}{2} \right) = n^2 + n$$

□

Observation 9: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=0}^{n-1} (2j+1) = n^2 \quad \text{or equivalently} \quad \sum_{j=1}^n (2j-1) = n^2$$

Proof. We have:

$$\begin{aligned} \sum_{j=0}^{n-1} (2j+1) &= 2 \sum_{j=0}^{n-1} j + \sum_{j=0}^{n-1} 1 = 2 \sum_{j=1}^n (j-1) + \sum_{j=1}^n 1 \\ &= 2 \left(\sum_{j=1}^n j - \sum_{j=1}^n 1 \right) + n \\ &= 2 \left(\frac{n^2+n}{2} - n \right) + n = n^2 + n - 2n + n = n^2 \end{aligned}$$

$$\begin{aligned} \sum_{j=1}^n (2j-1) &= 2 \sum_{j=1}^n j - \sum_{j=1}^n 1 \\ &= 2 \left(\frac{n^2+n}{2} \right) - n = n^2 + n - n = n^2 \end{aligned}$$

□